

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Indigo Bunting"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

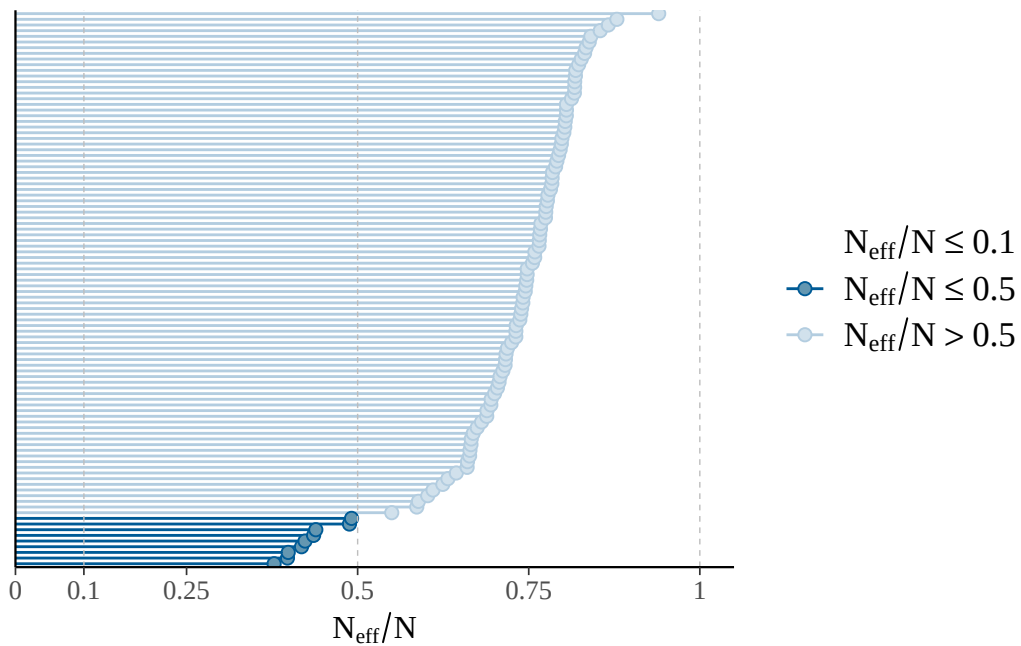
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3974	0.8585	1.0929	1.0890	1.3223	2.0482

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4389	0.6932	0.7473	0.7319	0.7965	0.9399

```
[1] "effective sample size in bulk and tail fine"
```

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.



Post-warmup iteration



Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9991	0.9995	0.9998	1.0000	1.0002	1.0032

```
[1] "All R-hat values < 1.01"
```

Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"      "b_occ_Ibio1_3yrsE2"  "b_occ_Ibio14_3yrsE2"
[4] "b_occ_secdf_3yrs"     "b_occ_urban_3yrs"    "b_occ_Isecdf_3yrsE2"
```

```
[1] "b_colo_bio1"          "b_colo_pr_summer"
[3] "b_colo_pr_winter"     "b_colo_Ibio1E2"
[5] "b_colo_Ipr_summerE2"  "b_colo_sum_annual_crops"
[7] "b_colo_secdf"         "b_colo_pastr"
[9] "b_colo_urban"         "b_colo_IpastrE2"
[11] "b_colo_IurbanE2"
```

```
[1] "b_ex_bio1"           "b_ex_Ibio1E2"
[3] "b_ex_Ibio2E2"        "b_ex_sum_annual_crops"
[5] "b_ex_secdf"          "b_ex_pastr"
[7] "b_ex_urban"          "b_ex_Isum_annual_cropsE2"
[9] "b_ex_IsecdfE2"       "b_ex_IpastrE2"
[11] "b_ex_IurbanE2"
```

